Strong Artificial Intelligence

G Memory

Persistencer

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Technical University of Applied Sciences, Würzburg, 29.05.2026

Repetition „Concept“

- What is a primitive, native and derived concept?
- What is the counterpart paradigm?
- What is not a concept?
- How can concepts be presented?
- Why is hourse not a hourse?
- How are concepts helping us to be curious?

Workshop „Memory“

- How and when do we remember experiences or impressions from the past?
- Which thoughts are coming from alone? And why?
- How is the access to persisted information implemented?
- Is persisted information information?
- How can we remember things that we will do in the future?
- In which parts or segments would you divide our memory?
- Which information would you put into which segment?

H Memory

1. Task	Persistencing and Recalling
2. Remembering	Associating
3. Forgetting	No Usage
4. Primitives	Name Time Space
5. Segments	Allocation of Knowledge
6. Implementation	Database
7. Conclusion	Core

1 Task

Persistencing and Recalling

thws Technische Hochschule Würzburg-Schweinfurt
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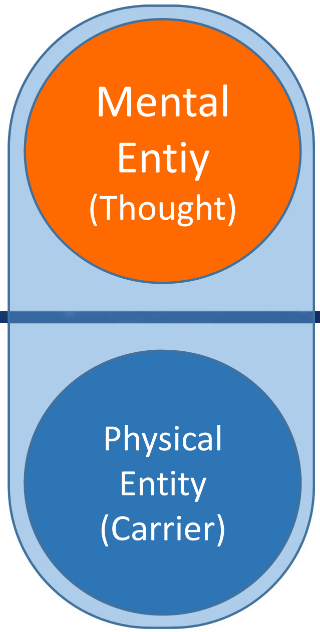
Persisting a thought

virtual world

mental world

thinking



physcial world

memory

Recalling a thought

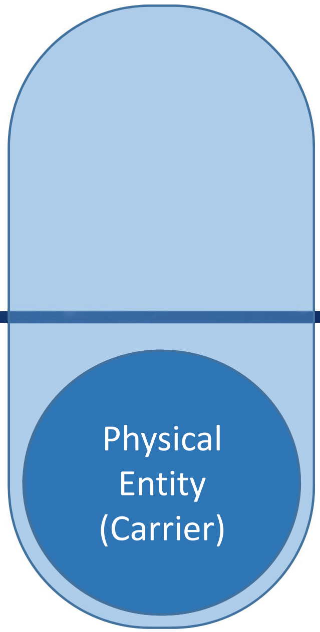
virtual world

mental world

thinking

physcial world

memory



1. Task

Primitive Concept Name and Name Association

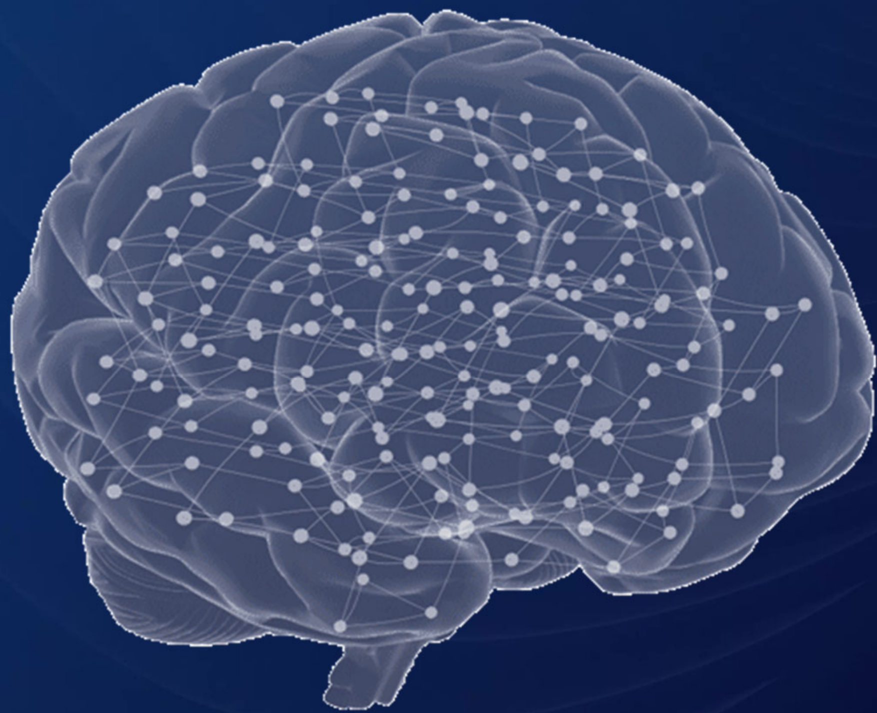
nominee — french name → La tour Eiffel

nominee — german name → Eiffelturm

nominee — english name → Eiffel Tower

1. Task

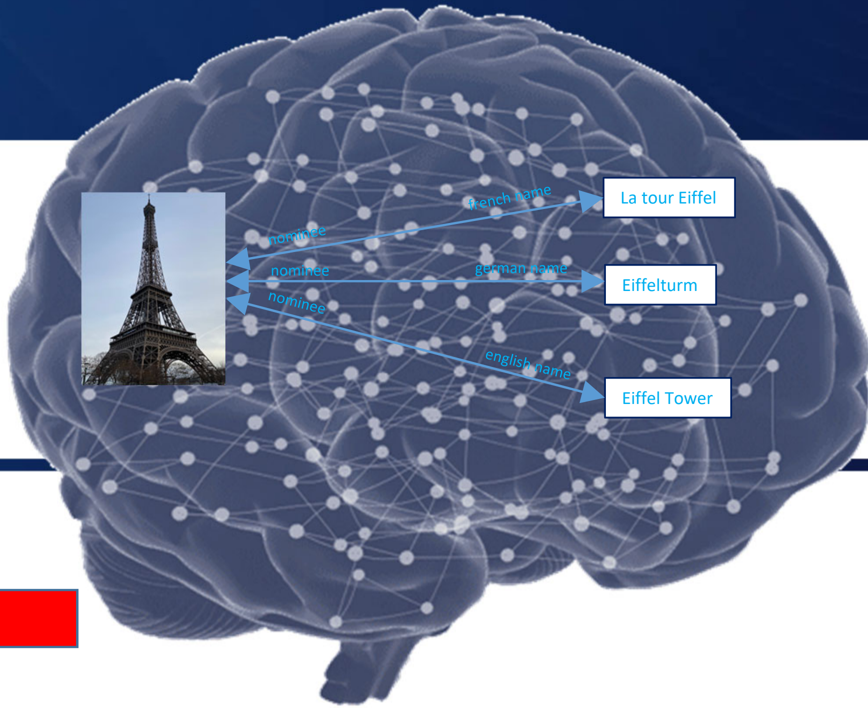
Persisting information



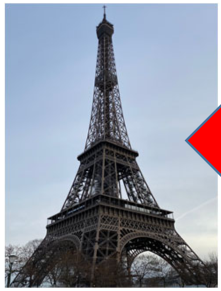
1. Task

Persisting information

mental world



physcial world



physical representation

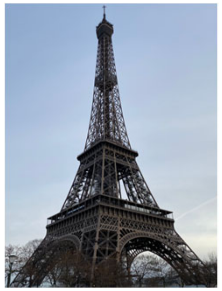
memory

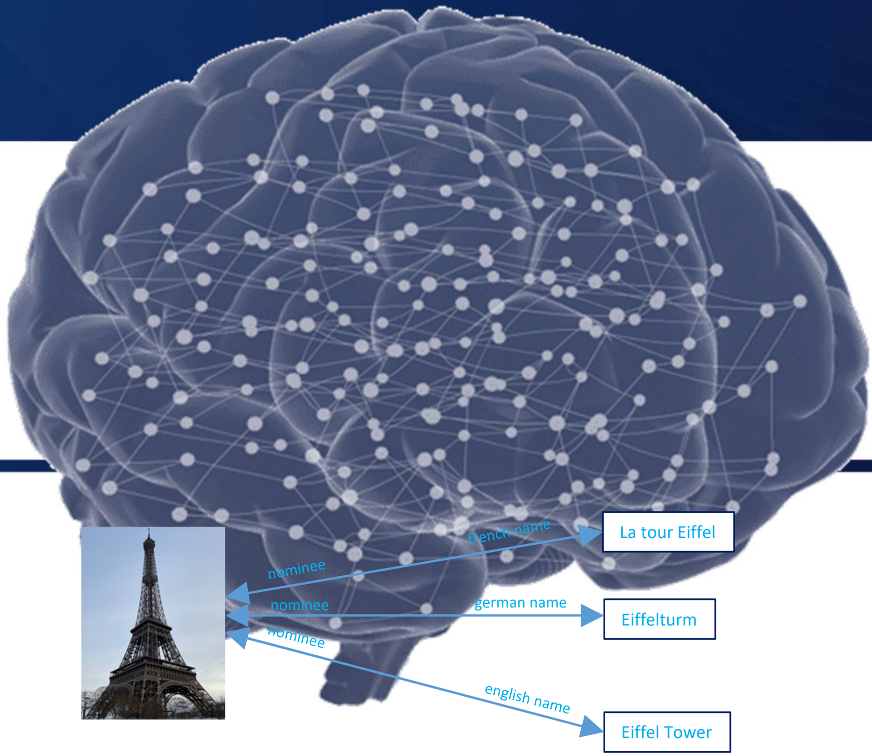
1. Task

Recall persisted information

mental world

physcial world





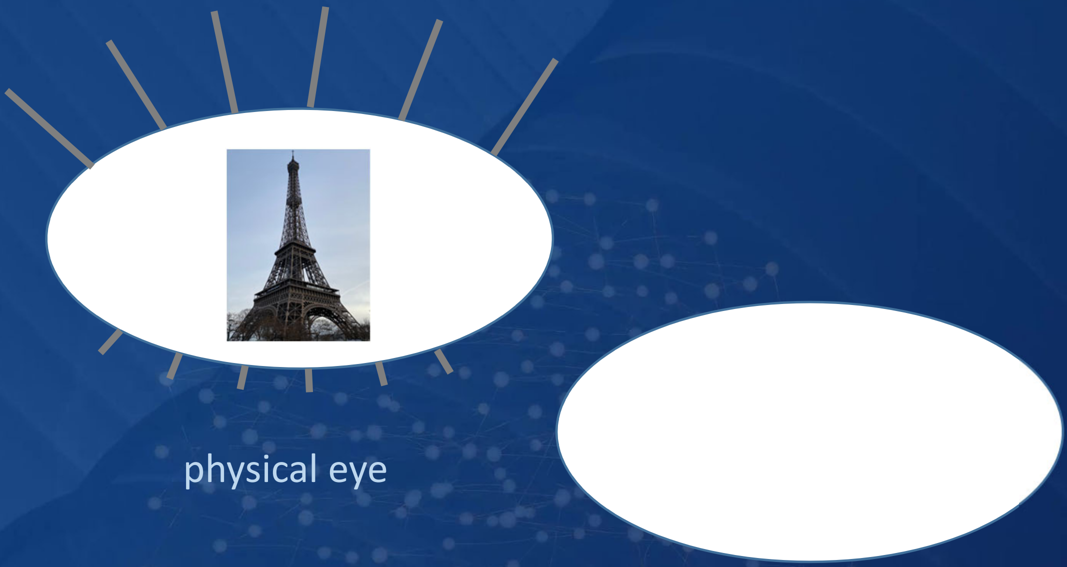
2 Remembering

Associating

Reasons for Remembering

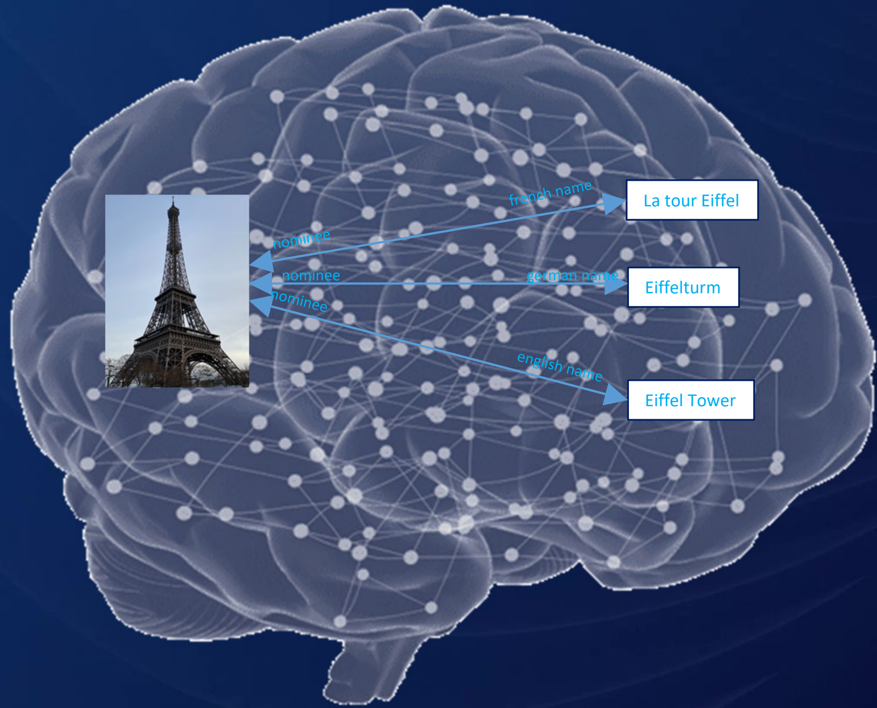
1. Impressions
   recalling persisted information on base of similar schemas
2. Uncomplete Concepts
   keep us thinking a lot to fill information gaps
3. Associative recall
   draws associated entites (memories) out of the memory
4. …

2. Remembering by Impressions

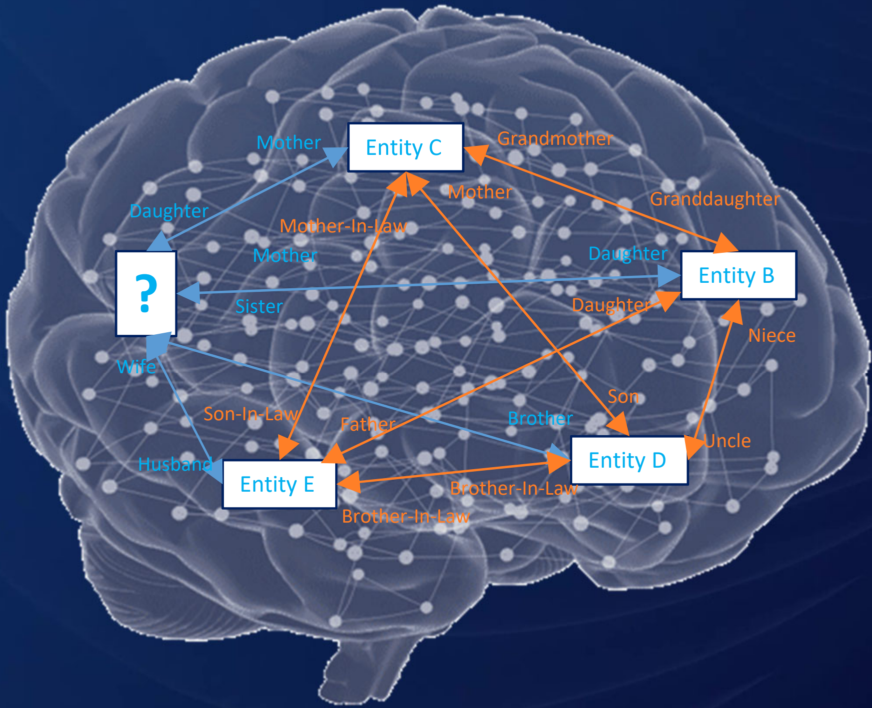


physical eye

mental eye



2. Remembering by Uncomplete Concepts



3 Forgetting

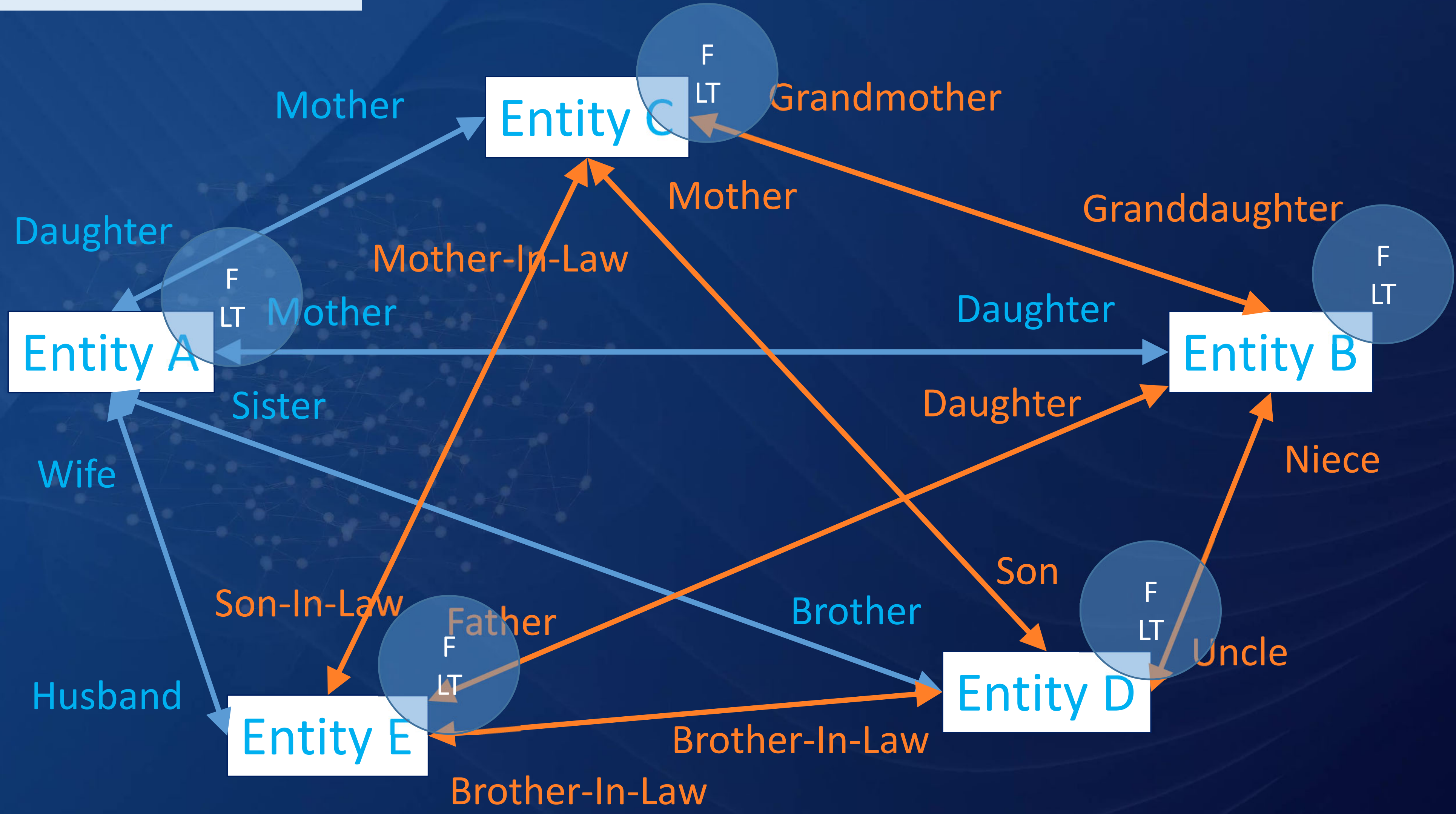
No Usage

Workshop „Forgetting“

- What do we forget and what not?
- How do we compensate for memory lapses?
- How would you implement forgetting information in a digital memory?

3. Forgetting

Frequency of Use (F) and Last Time of Use (LT)



4 Primitives

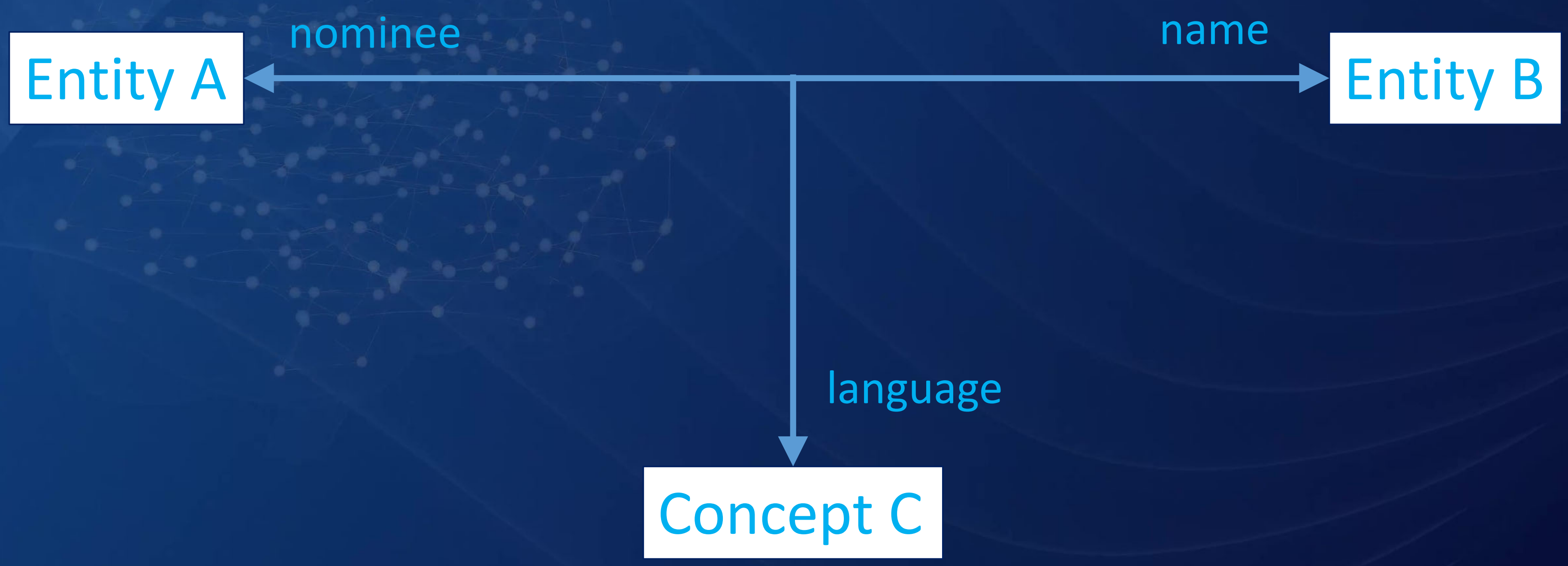
Name Time Space

4. Name

What is a Name?	When is it Used?
Phonetic/Textual Pattern.	When Individuals Communicate.
What is it Used for?	**Why is it Used?**
To Assign it to Mental Entities.	To Shortcut the communication.

4. Name

Simple Concept (Tertiary Association)



4. Name

Simple Concept

nominee

name

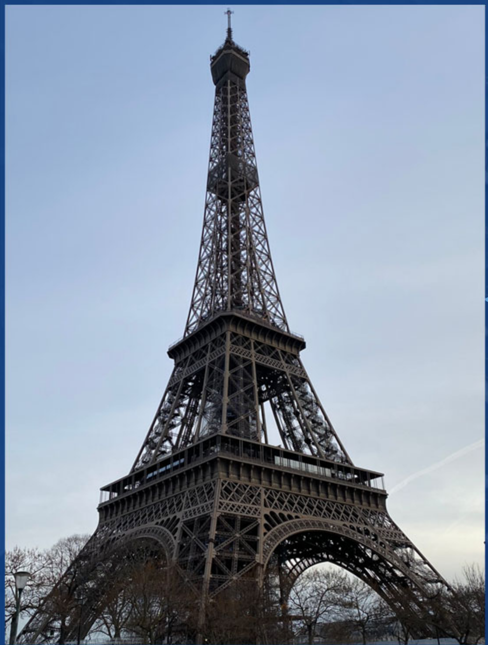
Eiffelturm

language

German

4. Name

Simple Concept



nominee

name

Eiffel Tower

language

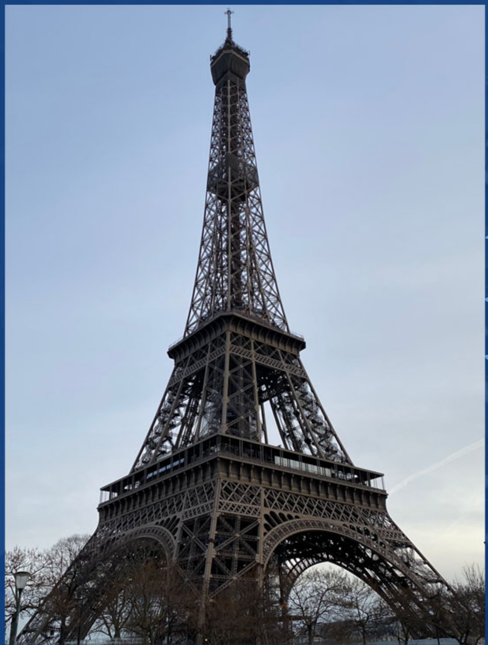
English

4. Name

Simple Concept (Binary Association)

4. Name

Simple Concept



nominee

french name

La tour Eiffel

nominee

german name

Eiffelturm

nominee

english name

Eiffel Tower

Workshop „Time and Space“

- How would you model time in an information network?
- How would you model space in an information network?

5 Segments

Allocation of Knowledge

5. Segments

Value Memory

- Our action theory assumes inner values to detect the necessity of action.
- Inner values must be persistet in our memory, otherwise we could not refer to them
- Inner values as part of our memory are always formed and deformed (developed) by impressions and experiences
- Guided impressions and experiences is called education
- Not the whole memory is representing inner values, the inner values must be separated or segmented from other memories
- inner values memories must be labeled or clearly identifiable

Workshop „Segments“

- What other segments of memory would you distinguish beside the value memory?
- How could several memory segments be connected together?

5. Segments

The informtion and knowledge for different task can be stored in different segments:

- Short-term Memory
- Long-term Memory
  1. Episode Information
  2. Reality Information (Primary, Secondary, …)
  3. Individual Reality Information
  4. Value Information
  5. Action Plans
  6. Motion Concepts
  7. Speech Concepts

6 Implementation

Database

Workshop „Implementation“

- Find out, what persisting systems are based on associative patterns.
- Which systems have intrinsic time logic, which have spatial logic?
- What classical categories of database management systems exist?
- Are databases the memory of the digital thought machine?

```
CREATE TABLE ENTITY (
        UUID CHAR (36) NOT NULL,
        IMPRINTTIME TIMESTAMP,
        IMPRESSOR CHAR (36),
        TYPE CHAR (2),
        NAME CHAR (36),
        PRIMARY KEY (UUID)
)

CREATE TABLE CONCEPT (
        UUID CHAR (36) NOT NULL,
        IMPRINTTIME TIMESTAMP,
        IMPRESSOR CHAR (36),
        NAME CHAR (36),
        PRIMARY KEY (UUID)
)

CREATE TABLE NAME (
        UUID CHAR (36) NOT NULL,
        IMPRINTTIME TIMESTAMP,
        IMPRESSOR CHAR (36),
        NAME VARCHAR (256),
        PRIMARY KEY (UUID)
)

CREATE TABLE ASSOCIATION (
        UUID CHAR(36) NOT NULL,
        IMPRINTTIME TIMESTAMP,
        IMPRESSOR CHAR(36),
        ENTITY1 CHAR (36),
        CONCEPT1 CHAR (36),
        CONCEPT2 CHAR (36),
        ENTITY2 CHAR (36),
        STARTTIME TIMESTAMP,
        ENDTIME TIMESTAMP,
        PRIMARY KEY (UUID)
)
```

7 Conclusion

Database

7. Conclusion

- Memory is crucial for intelligence.
- Memory is the central place where all information is stored and retreived.
- Associative organized access to persisted information has huge advantages.
- The model of the persisting system (memory system) is crucial.
- It must be complete in terms of time logic, space logic etc.
- Implementation can be done conventionally.

I Thank You